



Kali Linux installer menu (BIOS mode)

Graphical install

Install

Advanced options

>

Accessible dark contrast installer menu

>

Help

Install with speech synthesis





Software selection

At the moment, only the core of the system is installed. The default selections below will install Kali Linux with its standard desktop environment and the default tools.

You can customize it by choosing a different desktop environment or a different collection of tools.

Choose software to install:

- ☒ Desktop environment [selecting this item has no effect]
 - ☒ ... Xfce (Kali's default desktop environment)
 - ☐ ... GNOME
 - ☐ ... KDE Plasma
- ☒ Collection of tools [selecting this item has no effect]
 - ☒ ... top10 -- the 10 most popular tools
 - ☒ ... default -- recommended tools (available in the live system)
 - ☐ ... large -- default selection plus additional tools

Screenshot

Continue

NIDS → Network Intrusion Detection System:

A network Intrusion Detection System (NIDS) is a security tool that monitors network traffic to detect suspicious activities, attacks, or policy violations.

Key points →

- Works at the network level, not the device level.
- Analyzes packets moving across the network.
- Helps identify threats like malware, port scans, DoS attacks, suspicious traffic patterns, etc.
- Alerts security teams when abnormal behavior is detected.

Types of NIDS

1. Signature-based NIDS

- Detects attacks using known threat signatures (like antivirus).
- Accurate but cannot detect new/unknown attacks.

2. Anomaly-based NIDS

- Detects unusual behavior compared to normal network baseline.
- Can detect unknown attacks but may generate false alerts.

Snort

Snort is one of the most widely used open-source NIDS and intrusion prevention systems (IPS).

Who developed it?

- originally developed by martin roesch.
- now maintained by cisco.

Why Snort is popular?

- free and open-source.
- highly customizable.
- large community support.
- work on linux, windows and macos.

Snort operates in 3 main modes:

1. Packet Sniffer mode.

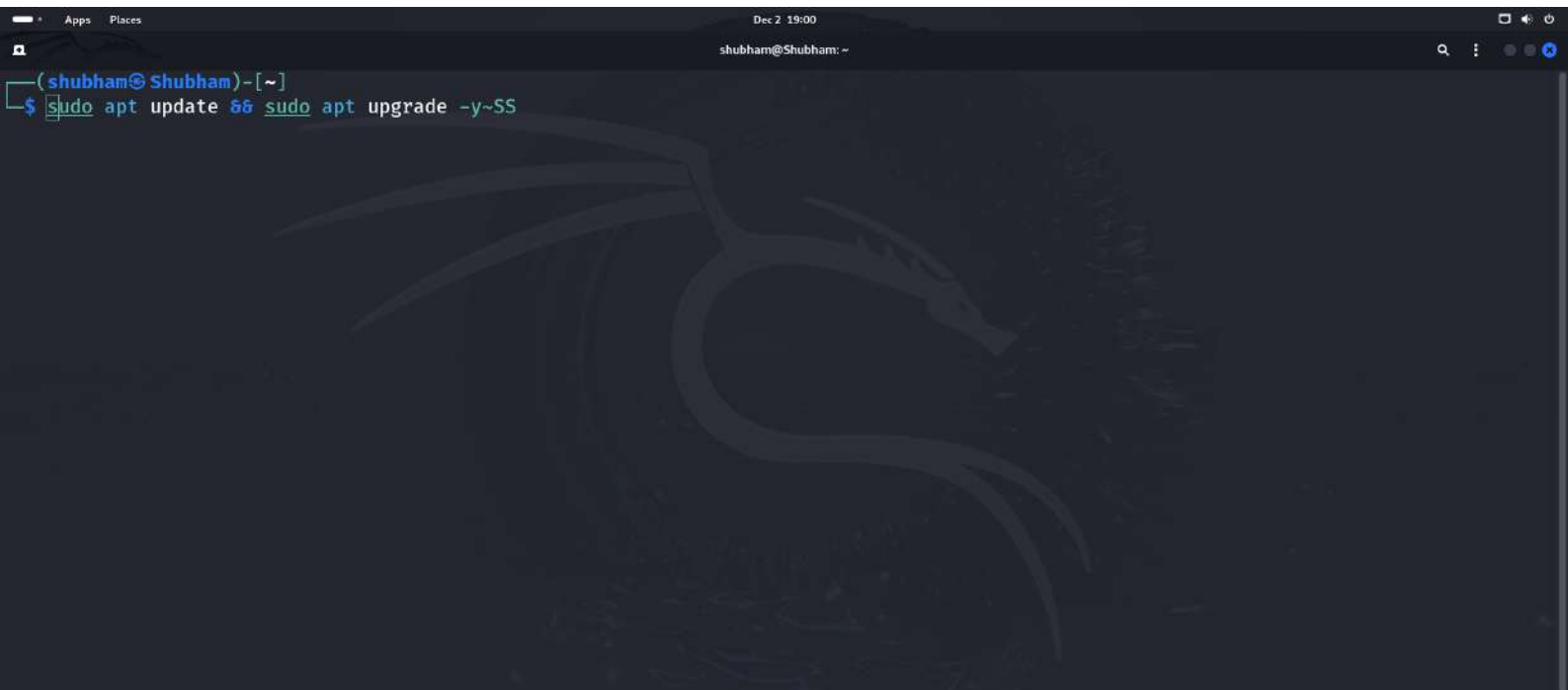
- captures and displays network packets in real time.

2. Packet Logger mode.

- Saves network packets to disk for later analysis.

3. NIDS mode.

- Uses rules to detect malicious traffic and generate alerts.




```
aspell-autobuildhash: processing: en [en_US-wo_accents-only].
Processing triggers for ca-certificates (20250419) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for initramfs-tools (0.150) ...
update-initramfs: Generating /boot/initrd.img-6.16.8+kali-amd64
Processing triggers for tex-common (6.19) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.4+dfsg-1) ...
Processing triggers for dbus (1.16.2-2) ...
Processing triggers for libc-bin (2.41-12) ...
Processing triggers for rygel (45.0-1) ...
Processing triggers for ca-certificates-java (20240118) ...
done.
```

```
Dec 2 20:09
shubham@Shubham: ~

(shubham@Shubham)-[~]
$ sudo apt install snort -y
[sudo] password for shubham:
The following packages were automatically installed and are no longer required:
amass-common libgail-common libjim0.82t64 python3-hatch-vcs
chrome-gnome-shell libgail18t64 libjsoncpp25 python3-hatchling
firebird3.0-common libgeos3.12.2 libmbedcrypto7t64 python3-kismetcapturebtgeiger
firebird3.0-common-doc libgfapi0 libmongoc-1.0-0t64 python3-kismetcapturefreaklabszigbee
firmware-ti-connectivity libgfrpc0 libnet1 python3-kismetcaptureertl433
fonts-liberation2 libgfxdr0 libpaper1 python3-kismetcaptureertladsb
freerdp2-x11 libgl1-mesa-dev libperl5.38t64 python3-kismetcaptureertlamr
hydra-gtk libglapi-mesa librados2 python3-packaging-whl
ibverbs-providers libgles-dev libradare2-5.0.0t64 python3-pathspect
icu-devtools libgles1 librdmacm1t64 python3-protobuf
libabsl20230802 libglusterfs0 libsframe1 python3-pysmi
libassuan0 libglvnd-core-dev libsigsegv2 python3-setuptools-scm
libbfio1 libglvnd-dev libtagc0 python3-trove-classifiers
libboost-iostreams1.83.0 libgtk2.0-0t64 libutempter0 python3-wheel-whl
libboost-thread1.83.0 libgtk2.0-bin libwinpr2-2t64 python3-zombie-imp
libbson-1.0-0t64 libgtk2.0-common libzip4t64 ruby-zeitwerk
libcapstone4 libgtksourceview-3.0-1 openjdk-17-jre ruby3.1
libcephfs2 libgtksourceview-3.0-common openjdk-17-jre-headless ruby3.1-dev
libconfig++9v5 libgtksourceviewmm-3.0-0v5 openjdk-23-jre ruby3.1-doc
libconfig9 libgumbo2 openjdk-23-jre-headless rwho
libegl-dev libibverbs1 perl-modules-5.38 rwhod
libfontconfig1 libinput-dev python3-urllib3
```

```
libfreerdp2-2t64      libinstpatch-1.0-2      python3-click-plugins
Use 'sudo apt autoremove' to remove them.

Installing:
snort

Installing dependencies:
libdaq3 libestr0 libfastjson4 liblognorm5 oinkmaster rsyslog snort-common snort-common-libraries snort-rules-default

Suggested packages:
rsyslog-doc      rsyslog-mongodb      rsyslog-hiredis      rsyslog-docker      | rsyslog-gnutls  snort-doc
rsyslog-mysql    rsyslog-elasticsearch rsyslog-snmp          rsyslog-clickhouse  rsyslog-gssapi
rsyslog-pgsql    rsyslog-kafka        rsyslog-kubernetes   rsyslog-openssl     rsyslog-relp

Summary:
Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 345
1 not fully installed or removed.
Download size: 4061 kB
Space needed: 17.6 MB / 16.1 GB available

Get:1 http://mirrors.esto.network/kali kali-rolling/main amd64 snort-common-libraries amd64 3.9.7.0-0kali2 [282 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 snort-rules-default all 3.9.7.0-0kali2 [232 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 snort-common all 3.9.7.0-0kali2 [129 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 libestr0 amd64 0.1.11-2 [9048 B]
Get:5 http://kali.download/kali kali-rolling/main amd64 libfastjson4 amd64 1.2304.0-2 [28.9 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 liblognorm5 amd64 3.9.7.0-1 [68.0 kB]
```



```
Dec 2 20:10
shubham@Shubham: ~

Preparing to unpack .../9-oinkmaster_2.0-5_all.deb ...
Unpacking oinkmaster (2.0-5) ...
Setting up oinkmaster (2.0-5) ...
Setting up libestr0:amd64 (0.1.11-2) ...
Setting up libfastjson4:amd64 (1.2304.0-2) ...
Setting up mitmproxy (12.2.0-0kali1) ...
File "/usr/lib/python3/dist-packages/mitmproxy/contentviews/_api.py", line 20
    type SyntaxHighlight = Literal["css", "javascript", "xml", "yaml", "none", "error"]
    ^^^^^^^^^^^^^^^^^^^^^
SyntaxError: invalid syntax
dpkg: error processing package mitmproxy (--configure):
 installed mitmproxy package post-installation script subprocess returned error exit status 1
Setting up snort-common (3.9.7.0-0kali2) ...
Setting up liblognorm5:amd64 (2.0.7-1) ...
Setting up libdaq3 (3.0.22-0kali2) ...
Setting up snort-rules-default (3.9.7.0-0kali2) ...
Setting up snort-common-libraries (3.9.7.0-0kali2) ...
Setting up rsyslog (8.2510.0-5) ...
Created symlink '/etc/systemd/system/syslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/rsyslog.service' → '/usr/lib/systemd/system/rsyslog.service'.
Setting up snort (3.9.7.0-0kali2) ...
snort.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.4.2) ...
Processing triggers for libc-bin (2.41-12) ...
Errors were encountered while processing:
```

```
Dec 2 20:11
shubham@Shubham: ~

Processing triggers for libc-bin (2.41-12) ...
Errors were encountered while processing:
mitmproxy
Error: Sub-process /usr/bin/dpkg returned an error code (1)

(shubham@Shubham)~$ snort -v
-----
o")~  Snort++ 3.9.7.0
-----
Network Policy : policy id 0 :
-----
Inspection Policy : policy id 0 :
-----
pcap DAQ configured to passive.
-----
host_cache
  memcap: 33554432 bytes

Snort successfully validated the configuration (with 0 warnings).
o")~  Snort exiting

(shubham@Shubham)~$
```

```
o")~ Snort exiting

(shubham@Shubham)~]
$ pwd
/home/shubham

(shubham@Shubham)~]
$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos new

(shubham@Shubham)~]
$ ls -l
total 36
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Desktop
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Documents
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Downloads
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Music
drwxr-xr-x 3 shubham shubham 4096 Dec 2 19:00 Pictures
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Public
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Templates
drwxr-xr-x 2 shubham shubham 4096 Nov 27 20:45 Videos
-rw-r--r-- 1 root root 1193 Dec 2 19:27 new

(shubham@Shubham)~]
$
```

```
(shubham@Shubham)-[~]
$ ls -a
.          .bashrc      .config     .java       .profile    .zshrc      Downloads  Public      new
..         .bashrc.original .face       .local      .sudo_as_admin_successful Desktop     Music      Templates
.bash_logout .cache      .face.icon  .mozilla    .zsh_history Documents   Pictures   Videos

(shubham@Shubham)-[~]
$ cd Documents

(shubham@Shubham)-[~/Documents]
$ cd ..

(shubham@Shubham)-[~]
$ cd ~

(shubham@Shubham)-[~]
$ cd /

(shubham@Shubham)-[/]
```

```
(shubham@Shubham)-[/]
$ nmap 192.168.1.14
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-02 21:02 IST
Nmap scan report for 192.168.1.14 (192.168.1.14)
Host is up (0.0000020s latency).
All 1000 scanned ports on 192.168.1.14 (192.168.1.14) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.09 seconds

(shubham@Shubham)-[/]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:b1:e7:c7 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.14/24 brd 192.168.1.255 scope global dynamic noprefixroute eth0
        valid_lft 80565sec preferred_lft 80565sec
    inet6 fe80::20c:29ff:feb1:e7c7/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```



```
Dec 2 21:09
shubham@Shubham: ~

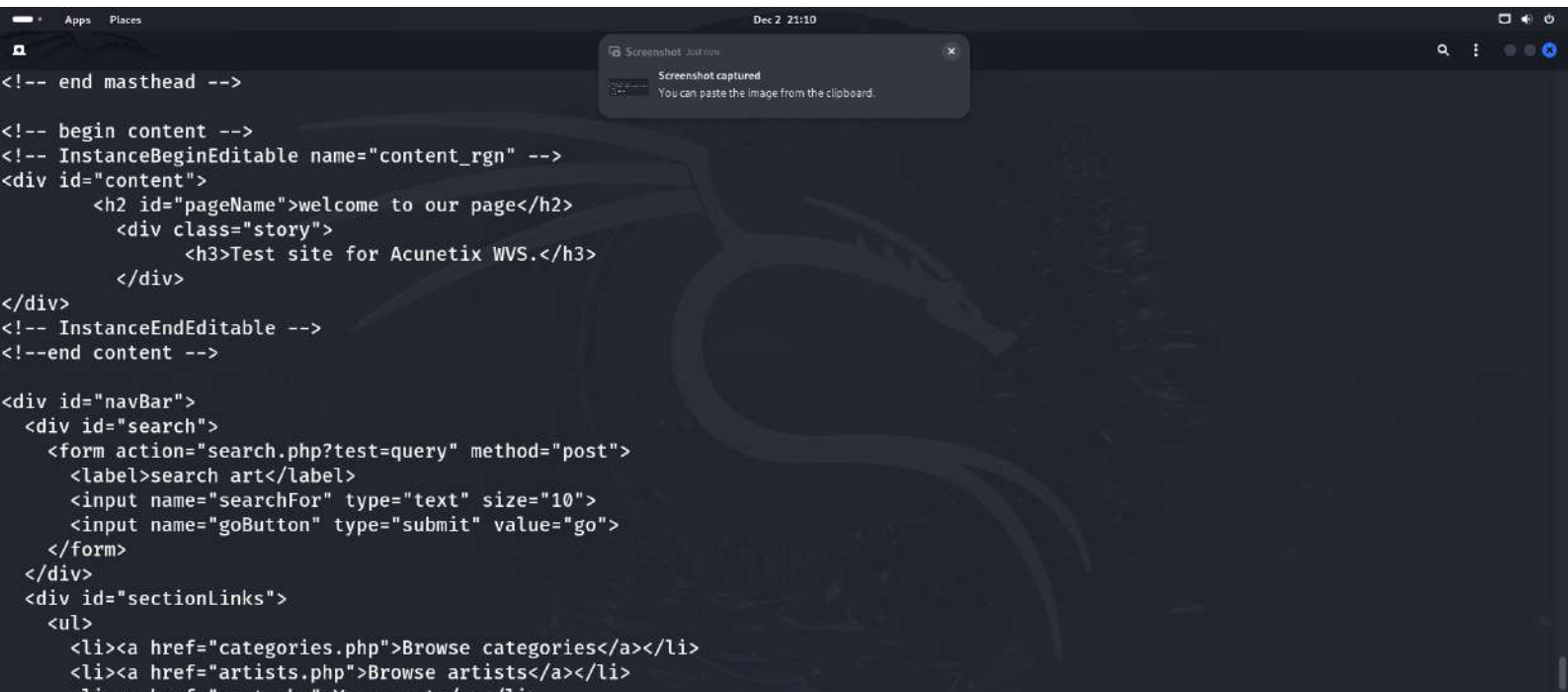
$ curl http://testphp.vulnweb.com
<!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN"
"http://www.w3.org/TR/html4/loose.dtd">
<html><!-- InstanceBegin template="/Templates/main_dynamic_template.dwt.php" codeOutsideHTMLOutsideIsLocked="false" -->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-2">

<!-- InstanceBeginEditable name="document_title_rgn" -->
<title>Home of Acunetix Art</title>
<!-- InstanceEndEditable -->
<link rel="stylesheet" href="style.css" type="text/css">
<!-- InstanceBeginEditable name="headers_rgn" -->
<!-- here goes headers headers -->
<!-- InstanceEndEditable -->
<script language="JavaScript" type="text/JavaScript">
<!--
function MM_reloadPage(init) { //reloads the window if Nav4 resized
  if (init==true) with (navigator) {if ((appName=="Netscape")&&(parseInt(appVersion)==4)) {
    document.MM_pgW=innerWidth; document.MM_pgH=innerHeight; onresize=MM_reloadPage; }}
  else if (innerWidth!=document.MM_pgW || innerHeight!=document.MM_pgH) location.reload();
}
MM_reloadPage(true);
//-->
</script>
```

```
Dec 2 21:10
shubham@Shubham: ~

//-->
</script>

</head>
<body>
<div id="mainLayer" style="position:absolute; width:700px; z-index:1">
<div id="masthead">
  <h1 id="siteName"><a href="https://www.acunetix.com/"></a></h1>
  <h6 id="siteInfo">TEST and Demonstration site for <a href="https://www.acunetix.com/vulnerability-scanner/">Acunetix Web Vulnerabilit
y Scanner</a></h6>
  <div id="globalNav">
    <table border="0" cellpadding="0" cellspacing="0" width="100%"><tr>
      <td align="left">
        <a href="index.php">home</a> | <a href="categories.php">categories</a> | <a href="artists.php">artists
        </a> | <a href="disclaimer.php">disclaimer</a> | <a href="cart.php">your cart</a> |
        <a href="guestbook.php">guestbook</a> |
        <a href="AJAX/index.php">AJAX Demo</a>
      </td>
      <td align="right">
      </td>
    </tr></table>
  </div>
</div>
<!-- end masthead -->
```



```
<!-- end masthead -->

<!-- begin content -->
<!-- InstanceBeginEditable name="content_rgn" -->
<div id="content">
    <h2 id="pageName">welcome to our page</h2>
    <div class="story">
        <h3>Test site for Acunetix WVS.</h3>
    </div>
</div>
<!-- InstanceEndEditable -->
<!--end content -->

<div id="navBar">
    <div id="search">
        <form action="search.php?test=query" method="post">
            <label>search art</label>
            <input name="searchFor" type="text" size="10">
            <input name="goButton" type="submit" value="go">
        </form>
    </div>
    <div id="sectionLinks">
        <ul>
            <li><a href="categories.php">Browse categories</a></li>
            <li><a href="artists.php">Browse artists</a></li>
            <li><a href="content.php">Your content</a></li>
        </ul>
    </div>
</div>
```

```
Dec 2 21:10
shubham@Shubham: ~

<li><a href="userinfo.php">Your profile</a></li>
<li><a href="guestbook.php">Our guestbook</a></li>
  <li><a href="AJAX/index.php">AJAX Demo</a></li>
</li>
</ul>
</div>
<div class="relatedLinks">
  <h3>Links</h3>
  <ul>
    <li><a href="http://www.acunetix.com">Security art</a></li>
    <li><a href="https://www.acunetix.com/vulnerability-scanner/php-security-scanner/">PHP scanner</a></li>
    <li><a href="https://www.acunetix.com/blog/articles/prevent-sql-injection-vulnerabilities-in-php-applications/">PHP vuln help
</a></li>
    <li><a href="http://www.electasy.com/Fractal-Explorer/index.html">Fractal Explorer</a></li>
  </ul>
</div>
<div id="advert">
  <p>
    <object classid="clsid:D27CDB6E-AE6D-11cf-96B8-444553540000" codebase="http://download.macromedia.com/pub/shockwave/cabs/flash/sw
    flash.cab#version=6,0,29,0" width="107" height="66">
      <param name="movie" value="Flash/add.swf">
      <param name="quality" value="high">
      <embed src="Flash/add.swf" quality="high" pluginspage="http://www.macromedia.com/shockwave/download/index.cgi?P1_Prod_Version=Sho
    ckwaveFlash" type="application/x-shockwave-flash" width="107" height="66"></embed>
    </object>
  </p>
</div>
```

```
Dec 2 21:10
shubham@Shubham: ~

<param name=quality value=high>
<embed src="Flash/add.swf" quality=high pluginspage="http://www.macromedia.com/shockwave/download/index.cgi?P1_Prod_Version=Sho
ckwaveFlash" type="application/x-shockwave-flash" width="107" height="66"></embed>
</object>
</p>
</div>
</div>

<!--end navbar -->
<div id="siteInfo"> <a href="http://www.acunetix.com">About Us</a> | <a href="privacy.php">Privacy Policy</a> | <a href="mailto:wvs@ac
unetix.com">Contact Us</a> | <a href="/Mod_Rewrite_Shop/">Shop</a> | <a href="/hpp/">HTTP Parameter Pollution</a> | &copy;2019
Acunetix Ltd
</div>

<br>
<div style="background-color:lightgray;width:100%;text-align:center;font-size:12px;padding:1px">
<p style="padding-left:5%;padding-right:5%"><b>Warning</b>: This is not a real shop. This is an example PHP application, which is inten
tionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad
configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip:
Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.</p>
</div>
</div>
</body>
<!-- InstanceEnd --></html>
```

```
Dec 2 21:10
shubham@Shubham: ~

</object>
</p>
</div>
</div>

<!--end navbar -->
<div id="siteInfo"> <a href="http://www.acunetix.com">About Us</a> | <a href="privacy.php">Privacy Policy</a> | <a href="mailto:wvs@ac
unetix.com">Contact Us</a> | <a href="/Mod_Rewrite_Shop/">Shop</a> | <a href="/hpp/">HTTP Parameter Pollution</a> | &copy;2019
Acunetix Ltd
</div>

<br>
<div style="background-color:lightgray;width:100%;text-align:center;font-size:12px;padding:1px">
<p style="padding-left:5%;padding-right:5%"><b>Warning</b>: This is not a real shop. This is an example PHP application, which is inten
tionally vulnerable to web attacks. It is intended to help you test Acunetix. It also helps you understand how developer errors and bad
configuration may let someone break into your website. You can use it to test other tools and your manual hacking skills as well. Tip:
Look for potential SQL Injections, Cross-site Scripting (XSS), and Cross-site Request Forgery (CSRF), and more.</p>
</div>
</div>
</body>
<!-- InstanceEnd --></html>

(shubham@Shubham) - [~]
$
```



```
Dec 2 21:14
shubham@Shubham: ~
64 bytes from 192.168.1.14: icmp_seq=129 ttl=64 time=0.054 ms
64 bytes from 192.168.1.14: icmp_seq=130 ttl=64 time=0.055 ms
64 bytes from 192.168.1.14: icmp_seq=131 ttl=64 time=0.055 ms
64 bytes from 192.168.1.14: icmp_seq=132 ttl=64 time=0.059 ms
64 bytes from 192.168.1.14: icmp_seq=133 ttl=64 time=0.056 ms
64 bytes from 192.168.1.14: icmp_seq=134 ttl=64 time=0.048 ms
64 bytes from 192.168.1.14: icmp_seq=135 ttl=64 time=0.077 ms
64 bytes from 192.168.1.14: icmp_seq=136 ttl=64 time=0.054 ms
64 bytes from 192.168.1.14: icmp_seq=137 ttl=64 time=0.057 ms
64 bytes from 192.168.1.14: icmp_seq=138 ttl=64 time=0.036 ms
64 bytes from 192.168.1.14: icmp_seq=139 ttl=64 time=0.042 ms
64 bytes from 192.168.1.14: icmp_seq=140 ttl=64 time=0.152 ms
64 bytes from 192.168.1.14: icmp_seq=141 ttl=64 time=0.032 ms
64 bytes from 192.168.1.14: icmp_seq=142 ttl=64 time=0.058 ms
64 bytes from 192.168.1.14: icmp_seq=143 ttl=64 time=0.130 ms
64 bytes from 192.168.1.14: icmp_seq=144 ttl=64 time=0.033 ms
64 bytes from 192.168.1.14: icmp_seq=145 ttl=64 time=0.058 ms
64 bytes from 192.168.1.14: icmp_seq=146 ttl=64 time=0.055 ms
^C
--- 192.168.1.14 ping statistics ---
146 packets transmitted, 146 received, 0% packet loss, time 147837ms
rtt min/avg/max/mdev = 0.028/0.060/1.486/0.119 ms
(shubham@Shubham) - [~]
$
```